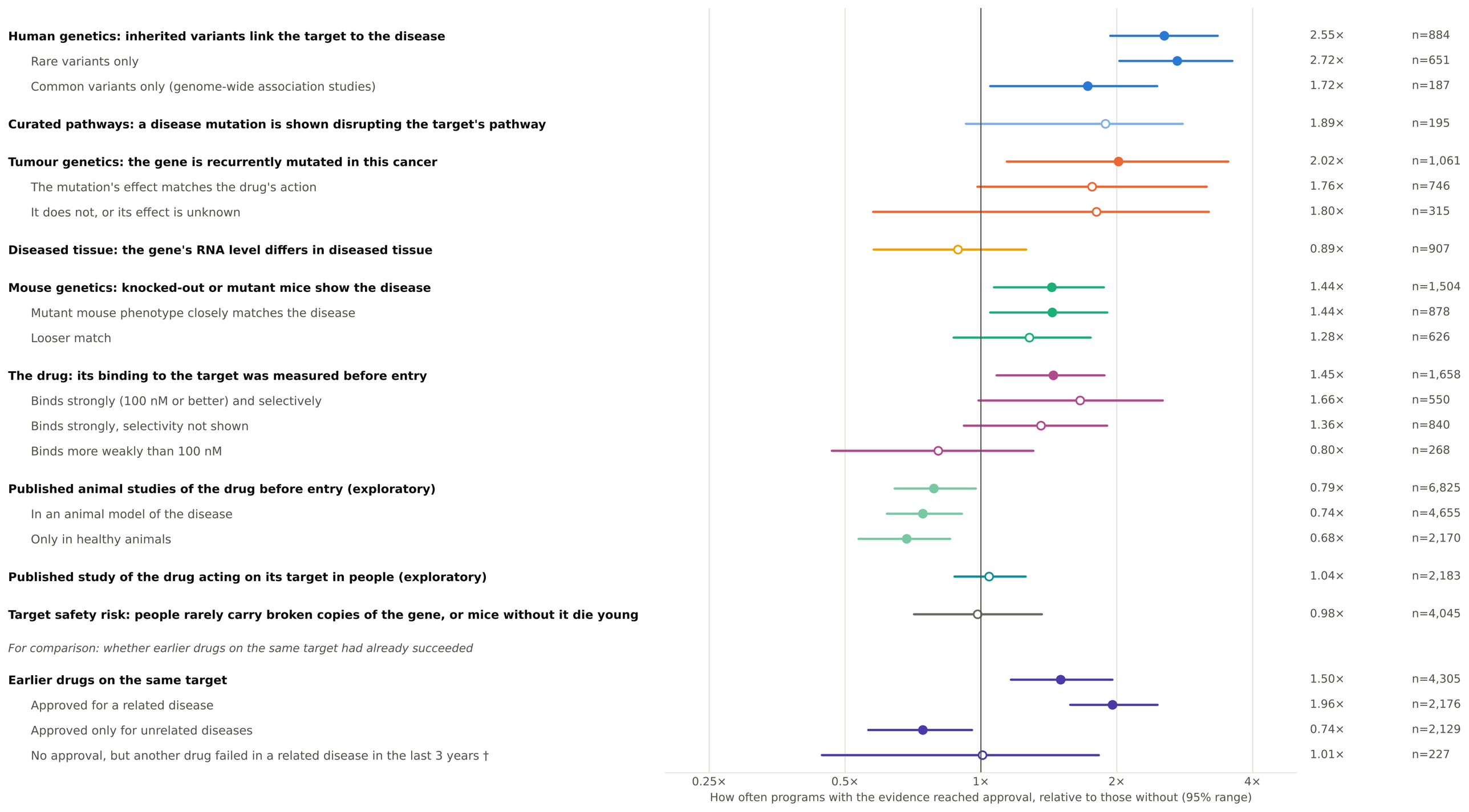
Human genetics: inherited variants link the target to the disease
2.55× n=884
Rare variants only
2.72× n=651
Common variants only (genome-wide association studies)
1.72× n=187
Curated pathways: a disease mutation is shown disrupting the target's pathway
1.89× n=195
Tumour genetics: the gene is recurrently mutated in this cancer
2.02× n=1,061
The mutation's effect matches the drug's action
1.76× n=746
It does not, or its effect is unknown
1.80× n=315
Diseased tissue: the gene's RNA level differs in diseased tissue
0.89× n=907
Mouse genetics: knocked-out or mutant mice show the disease
1.44× n=1,504
Mutant mouse phenotype closely matches the disease
1.44× n=878
Looser match
1.28× n=626
The drug: its binding to the target was measured before entry
1.45× n=1,658
Binds strongly (100 nM or better) and selectively
1.66× n=550
Binds strongly, selectivity not shown
1.36× n=840
Binds more weakly than 100 nM
0.80× n=268
Published animal studies of the drug before entry (exploratory)
0.79× n=6,825
In an animal model of the disease
0.74× n=4,655
Only in healthy animals
0.68× n=2,170
Published study of the drug acting on its target in people (exploratory)
1.04× n=2,183
Target safety risk: people rarely carry broken copies of the gene, or mice without it die young
0.98× n=4,045
For comparison: whether earlier drugs on the same target had already succeeded
Earlier drugs on the same target
1.50× n=4,305
Approved for a related disease
1.96× n=2,176
Approved only for unrelated diseases
0.74× n=2,129
No approval, but another drug failed in a related disease in the last 3 years †
1.01× n=227
0.25×
0.5×
1×
2×
4×
How often programs with the evidence reached approval, relative to those without (95% range)